

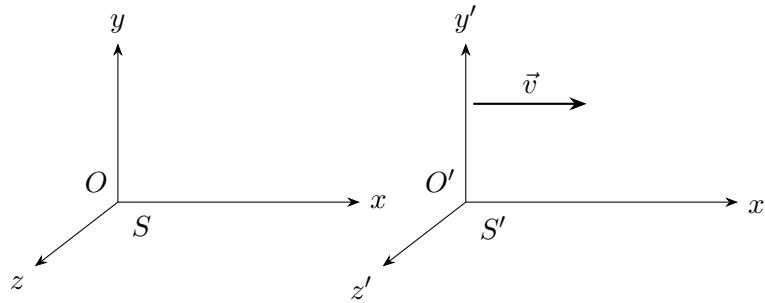
Special Relativity Notes

Derivation of the Lorentz Transformation, Velocity Transformation and Invariance of the Interval

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Setting up the Derivation

S' is moving with a velocity $\vec{v}$ with respect to S along the x -direction.



$$x' = x - vt$$

$$x' = K(x - vt)$$

$$x = x' + vt'$$

$$x = K(x' + vt')$$

Objective: To find K , and to find the relation between t' and t .

Finding K

Substituting x' into the second equation:

$$x = K(K(x - vt) + vt')$$

$$x = K^2(x - vt) + vt'K$$

Solving for t' (in terms of x and t):

$$t' = \frac{x}{Kv} (1 - K^2) + Kt$$

Thought experiment (lightning): a light signal has $x = ct$ and $x' = ct'$. Then

$$x' = ct' \Rightarrow K(x - vt) = ct'$$

$$\Rightarrow x = \underbrace{\left[\frac{cK + Kv}{K - \frac{c}{Kv}(1 - K^2)} \right]}_{=c} t$$

(since $x = ct$, the bracket must equal c).

$$\begin{aligned} \Rightarrow K^2 v^2 &= c^2 (K^2 - 1) \\ \Rightarrow K^2 &= \frac{c^2}{c^2 - v^2} \end{aligned}$$

$$K = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Finding t'

Route 1

$$\begin{aligned} t' &= \frac{x - K^2(x - vt)}{Kv} \\ &= \frac{x - K^2x + K^2vt}{Kv} \\ &= \frac{x(1 - K^2) + K^2vt}{Kv} \\ &= \frac{K^2vt - x(K^2 - 1)}{Kv} \\ &= \frac{K^2vt - xK^2\frac{v^2}{c^2}}{Kv} \quad \left(\because K^2 - 1 = K^2\frac{v^2}{c^2} \right) \\ &= \frac{t - \frac{xv}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

Route 2 (substituting K directly)

$$\begin{aligned} t' &= \frac{x - K^2(x - vt)}{Kv}, \quad K = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{x - \frac{x - vt}{1 - \frac{v^2}{c^2}}}{\frac{v}{\sqrt{1 - \frac{v^2}{c^2}}}} = \frac{[x(1 - \frac{v^2}{c^2}) - x + vt]}{v\sqrt{1 - \frac{v^2}{c^2}}} = \frac{vt - \frac{xv^2}{c^2}}{v\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

$$\therefore t' = \frac{t - \frac{xv}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

1 Velocity Transformation

$$U_x = \frac{dx}{dt}, \quad U'_x = \frac{dx'}{dt'}$$

$U \rightarrow$ velocity with respect to the rest frame, $U' \rightarrow$ velocity with respect to the moving frame.

$$U'_x = \frac{dx'/dt}{dt'/dt}$$

$$\frac{dx'}{dt} = \frac{U_x - v}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad \frac{dt'}{dt} = \frac{1 - \frac{v}{c^2}U_x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$U'_x = \frac{U_x - v}{1 - \frac{v}{c^2}U_x}$$

$$U'_y = \frac{dy'}{dt'}, \quad U_y = \frac{dy}{dt}; \quad U'_z = \frac{dz'}{dt'}, \quad U_z = \frac{dz}{dt}$$

$$U'_y = \frac{dy/dt}{dt'/dt}$$

$$U'_y = \frac{U_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2}U_x}$$

$$U'_z = \frac{U_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2}U_x}$$

Forward and inverse transformations

- **Forward transformation** ($S \rightarrow S'$): S' moves with $+v$ relative to S .
- **Inverse transformation** ($S' \rightarrow S$): from the viewpoint of S' , S moves with velocity $-v$.

Replace $v \rightarrow -v$:

$$x = K(x' + vt'), \quad t = K\left(t' + \frac{vx'}{c^2}\right)$$

$$U_x = \frac{U'_x + v}{1 + \frac{v}{c^2}U'_x}$$

$$U_y = \frac{U'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c^2}U'_x}$$

$$U_z = \frac{U'_z \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c^2}U'_x}$$

Einstein velocity addition rule

$$V_{AC} = \frac{V_{AB} + V_{BC}}{1 + \frac{V_{AB} V_{BC}}{c^2}}$$

2 Invariance of the Interval (Cartesian Coordinates)

$$\begin{aligned} ds^2 &= dx^2 + dy^2 + dz^2 - c^2 dt^2 \\ ds'^2 &= dx'^2 + dy'^2 + dz'^2 - c^2 dt'^2 \end{aligned}$$

or

$$\begin{aligned} s^2 &= x^2 + y^2 + z^2 - c^2 t^2 \\ s'^2 &= x'^2 + y'^2 + z'^2 - c^2 t'^2 \\ &= K^2(x - vt)^2 + y^2 + z^2 - c^2 K^2 \left(t - \frac{vx}{c^2}\right)^2 \\ &= K^2 \left[(x - vt)^2 - c^2 \left(t - \frac{vx}{c^2}\right)^2 \right] + y^2 + z^2 \\ &= K^2 \left[x^2 + v^2 t^2 - 2vtx - c^2 t^2 + 2vtx - \frac{v^2 x^2}{c^2} \right] + y^2 + z^2 \\ &= K^2 \left[x^2 \left(1 - \frac{v^2}{c^2}\right) + t^2(v^2 - c^2) \right] + y^2 + z^2 \end{aligned}$$

Using $v^2 - c^2 = -c^2 \left(1 - \frac{v^2}{c^2}\right)$:

$$\begin{aligned} s'^2 &= K^2 \left[x^2 \left(1 - \frac{v^2}{c^2}\right) - c^2 t^2 \left(1 - \frac{v^2}{c^2}\right) \right] + y^2 + z^2 \\ &= x^2 - c^2 t^2 + y^2 + z^2 \\ &= x^2 + y^2 + z^2 - c^2 t^2 \\ &= s^2 \end{aligned}$$

$\therefore s^2$ is **invariant**.